Multilevel Bayesian Modelling

Recorded Arrest Associations

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1 Abstract

This academic group study uses Bayesian logistic regression to describe associations in recorded stop-and-arrest data. It compares fixed effects with precinct-specific intercepts and time-of-day slopes, then adds a recorded weapon indicator. The supplied archive does not establish individual ownership or the complete team list; no sole-authorship claim is made.

The source specifies four chains per candidate, 3,000 iterations per chain and 1,500 warm-up iterations, with informative fixed-effect priors attributed to a separate 2024 analysis. A successful run would target 6,000 retained draws per candidate. This is a configuration count, not a verified draw count or effective sample size.

The 2025 workbook, historical prior fit, posterior objects and LOO diagnostics are unavailable. This report is therefore a specification and code audit, with no regenerated coefficient, performance, convergence or winning-model claim. The review replaces an unsafe time parser with explicit HH:MM/HHMM handling and supplies nine base-R contract examples; those R checks were not executed in this environment.

The research decision is to restore the complete data/fit workflow and validate preprocessing and predictive diagnostics before interpreting associations. The intended scope is descriptive population-level analysis, not an individual criminality score, a causal demographic effect or a policing-fairness verdict.

2 Recorded Population and Preprocessing

The script reads a 2025 workbook, restricts records to January–March, recodes the arrest indicator and constructs demographic, age, precinct, weapon and time variables. It removes records missing required model variables and restricts recorded ages to ten through ninety. Age bands are at most seventeen, eighteen–twenty-four, twenty-five–thirty-four, thirty-five–forty-four and forty-five or above. Several smaller recorded race categories are collapsed into an OTHER factor.

The source's original hour extraction takes the first two characters. It can interpret an integer HHMM value such as 930 as hour 93, or reject an unpadded time such as 9:30. The revised function explicitly accepts colon-separated HH:MM and one-to-four-digit HHMM, validates hours zero–23 and minutes zero–59 and returns missing for unsupported encodings. For example, both 930 and 09:30 map to hour nine, while 30 maps to 00:30 under the stated HHMM convention.

The actual workbook's time storage must still be inspected. Excel date-time fractions, dates and other formats must not be guessed into a clock convention. Sample counts, exclusions, category frequencies and the time parser's effect cannot be verified without that input. Complete-case restriction also changes the target population and may create selection bias.

3 Three Model Specifications

For recorded arrest outcome Y_i, each model assumes

$$Y_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \eta_i. \tag{1}$$

Table 1: Candidate structure in the supplied source.

Candidate	Fixed predictors	Precinct structure
1	Race, sex, age band, time	None
2	Same fixed predictors	Intercept and time slopes
3	Same plus weapon indicator	Intercept and time slopes

In the hierarchical candidates, a precinct can have its own baseline and time-of-day adjustments. The double-bar syntax in the source specifies uncorrelated group effects, rather than estimating their full covariance matrix. Partial pooling can stabilise sparse precinct estimates, but the strength of pooling must be assessed from an actual posterior.

The full candidate's weapon indicator may be observed during or after the process associated with arrest. Without a timestamped information definition, it must not be interpreted as a prospective predictor available at a decision point. Comparing descriptive conditional fits is different from building a real-time scoring service.

4 Priors, Predictive Comparison and Diagnostics

Fixed-effect Normal priors are attributed to a separate Q1 2024 fit. Precinct standard-deviation priors are lognormal. The missing 2024 workflow prevents checking how those priors were estimated, how the prior sample differs or whether the parameter names match the final design matrix. Prior sensitivity and prior predictive checks are prerequisites to a strong interpretation.

The source calls leave-one-out predictive comparison for all three models and posterior predictive checks for the full candidate. Saved LOO differences, standard errors and Pareto-k diagnostics are absent, so the full model is not called selected merely because later plotting uses it. A previously removed legacy DIC approximation is not reinstated.

Record-wise LOO usually conditions on precincts represented by other observations. It does not test generalisation to a new precinct or a later quarter. A suitable evaluation must define that target and use grouped or chronological boundaries accordingly. R-hat, effective sample sizes, divergences and tree-depth limits require the actual sampler outputs; specified tuning settings do not establish convergence.

5 Audit Results and Reproduction Status

Table 2: What the available evidence establishes.

Item	Status
Model source and configured prior values	Available
Exact 2025 workbook / 2024 prior workflow	Missing
Posterior and predictive artifacts	Missing
LOO comparison and Pareto diagnostics	Missing
Time-parser examples	Written; not executed in R
Reproduced model results	None claimed

The new parser contracts include padded and unpadded HHMM, colon times, midnight, the final valid minute and invalid-hour/minute cases. The base-R check script can be run independently of Stan once an R runtime is available. This review did not execute R or any Bayesian fit. Source review alone cannot confirm posterior computation or performance.

The correction is a preprocessing safeguard, not a claim that the missing workbook contains a particular representation or a quantified number of affected rows. The original fit environment, prior code and processed population must be restored before result-level claims are made.

6 Interpretation and Next Milestone

An association among records already selected into an administrative process does not identify population behaviour or the causal effect of a demographic characteristic. Recording practices, case selection, missingness, measurement error and precinct composition can all affect the fitted associations. A predictive comparison is not itself a substantive fairness assessment.

The next milestone is to recover the exact 2024 and 2025 snapshots and prior workflow, record preprocessing exclusions, validate factor/parameter names and fit one candidate with complete diagnostics. Then compare all three candidates using the declared prediction target, inspect difficult LOO observations and report posterior predictive checks. Until that succeeds, the project is a model-specification example rather than a reproduced numerical showcase.

References

[1] Project authors. Pinned source code and evidence. Original study distinguished from the November 2026 audit.

[2] Companion verification record. Seed, inputs, numerical scope and limitations.